Noise Reduction Method Based on Principal Component Analysis With Beta Process for Micro-Doppler Radar Signatures

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***Abstract*—In radar remote-sensing area, the radar returns from a target are usually under relatively low signal-noise ratio (SNR) due to the large distance between radar and target, which will bring difficulties in target detection, tracking, and classification. In this paper, an efficient algorithm is proposed to denoise the returned micro-Doppler radar signals under low SNR conditions. The new algorithm develops a nonparametric extension to the principal component analysis (PCA) model with the Beta process (BP) prior. The BP is a fully Bayesian conjugate prior which allows analytic posterior calculation and straightforward interference. This proposed Beta process-based principal component analysis (BP-PCA) is utilized to model the returned micro-Doppler signals from airplane targets and ground moving targets with low-resolution radar, where the number of principal components in PCA can be selected adaptively with the BP prior-based Bayesian structure. Noise reduction is accomplished via reconstructing the echo within the subspace that composed of the selected principal components and discarding the residual noise subspace. We demonstrate the noise reduction performance of the proposed model with measured micro-Doppler data from some different kinds of targets. The experimental results are also compared with some other state-of-the-art approaches.**

***Index Terms*—Hierarchical Bayes, low-resolution radar, micro-Doppler effect, noise reduction, principal component analysis (PCA).**

I. Introduction

When radar illuminates a moving target, the frequency of the received signal will be shifted from that of the transmitted signal, which is known as the Doppler effect [1]. In many cases, besides the bulk motion, a target or any structural component on it may have mechanical vibration or rotation motion, such as the swinging arms and legs of a walking human, the rotating rotor blades of a helicopter, the precession of space cone-shaped target, and so on. These motions can be called micromotion and such micromotion may induce additional frequency modulations around the target's Doppler frequency caused by the bulk motion in the Doppler domain [2], [4], [7]. This phenomenon is referred to as micro-Doppler effect [3]. Micromotion and corresponding micro-Doppler effect are highly related to the property of the target which can provide us more detailed information about the target [5], [8], [20], [22].

Since the micro-Doppler effect was first introduced in coherent laser radar systems by Chen *et al.*, there have been lots of researches on the application of micro-Doppler effects these years in radar area [6], [9]–[12], [14], [16], [18], [24], of which recent paper [6] presents a review of the work published in the last 10 years on emerging applications of radar target analysis using micro-Doppler signatures. Papers [11] and [18] investigate the micro-Doppler signatures in ISAR/SAR image. Papers [12], [15], and [21] analyze the time-varying micro-Doppler signatures for moving vehicles and paper [16] extracts the micro-Doppler features to categorize the moving vehicles into two kinds, i.e., 1) the wheeled vehicles and 2) tracked vehicles. Micro-Doppler signatures generated by the rotating blades on the airplanes targets are referred to as jet engine modulation (JEM), which are analyzed in papers [17] and [19]; papers [5] and [22] extract the micro-Doppler signatures to realize the classification of different kinds of airplane targets. In micro-Doppler study, human motion is another important topic. Feature extraction and pattern classification for different activities of human individuals are discussed in [9], [10], [13], and [23]. Paper [10] proposes a novel feature extraction method for micro-Doppler signatures and realizes the real-time process based on low-cost embedded processing devices, where the robust principal component analysis (RPCA) method is used for the feature dimensionality reduction. Paper [13] proposes a translation and rotation invariant micro-Doppler feature extraction method based on the use of the pseudo-Zernike moments. The above-mentioned papers have demonstrated that the micro-Doppler signatures do provide us more detailed information about the target and the corresponding micro-Doppler features can be utilized for the classification of targets. Papers [5], [16], and [22] demonstrate that the experimental classification performance deteriorates dramatically in the low SNR cases, since in the low SNR cases, the micromotion components are contaminated by the noise and the noised micro-Doppler signatures further affect the parameter estimation or feature extraction in the classification methods. Thus, some preprocessing methods are needed to denoise the micro-Doppler signatures under the low SNR condition. However, to the best of our knowledge, denoising methods for micro-Doppler signals are not considered in the existing work except for the method proposed in [5],

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which requires a precise estimation of the noise variance. In this paper, we propose a novel noise reduction method based on principal component analysis (PCA), which does not need the prior information of the noise and is more effective than the method in [5].

PCA is an important data analysis method in statistic pattern recognition which can decompose high-dimensional data into a low-dimensional signal subspace and a noise subspace [25], [26]. This decomposition can minimize the mean square error (MSE) between the data and their projections by discarding the noise subspace. Therefore, data compression and denoising can be accomplished. In many applications, PCA is a useful first step for the data-processing tasks. The size of signal subspace reflects the amount of information contained in data. If the estimated size of signal subspace is smaller than the real one, some useful information will be lost. On the contrary, if that is larger than the real one, some fake information will be included. Thus, the precise estimation of the size of signal subspace is the central issue in PCA. In this regard, there are some threshold-based schemes, e.g., setting a threshold for the ratio of each eigenvalue to the total power, proposed to separate the eigenvalues into signal and noise subspaces [27]. However, these methods require a user-defined threshold and their size estimation results are not precise in most cases. After the probability model of PCA, i.e., probability principal component analysis (PPCA), is proposed by Tipping and Bishop in [28], papers [29] and [30], respectively, apply the Akaike's Information Criterion (AIC) [31] and Bayesian Information Criterion (BIC) [32] to the real-valued PPCA model to estimate the size of the signal subspace. However, as a parametric approach, such a model selection method needs to search all the possibilities of subspace dimension, which is time-consuming especially for the high-dimensional signals.

The research reported here proposes a Bayesian nonparametric model to the PCA problem via a Beta process (BP) for the purpose of denoising. This Beta process principal component analysis (BP-PCA) model can adaptively choose the number of principal components without the knowledge of the noise variance. We derive a variational Bayesian (VB) inference algorithm for the BP-PCA model and demonstrate that this model can work well on the measured data from airplane targets and ground moving targets with low-resolution radar.

The main contributions of this paper are summarized as follows. 1) We propose a nonparametric extension to the PCA model with the BP prior, which leads to a novel Beta process-based principal component analysis model, i.e., BP-PCA. 2) We derive an efficient VB algorithm to obtain the analytic posterior estimation and straightforward interference for the proposed BP-PCA model.

This paper is organized as follows. In Section II, we introduce the proposed BP-PCA model and derive the VB inference algorithm with a simulation example to verify its effectiveness. In Section III, based on the analysis of the measured micro-Doppler signatures of airplane targets and ground moving targets with low-resolution radar, we apply the proposed denoising method to the measured data to evaluate its noise reduction performance with comparison with some related noise reduction methods. Finally, we summarize this paper in Section IV.

II. PCA With BP

A. BP-Based PCA

Since the radar receiver adopts the quadrature sampling technique, the radar signal can be regarded as a complex signal consisting of the inphase and quadrature components. Thus, in the following discussion, we focus on the denoising problem of complex signals.

Consider a data set of time-domain observations $\{\boldsymbol{x}_n\}$ with $n = 1, \ldots, N$ and $\boldsymbol{x}_n \in \mathbb{C}^{D\times 1}$, where $\mathbb{C}^{D\times 1}$ denotes $\boldsymbol{x}_n$ being a complex vector of length D.

PCA seeks to project the observations to the subspace spanned by the M ($M < D$) eigenvectors corresponding to the M largest eigenvalues of the observation covariance matrix $\boldsymbol{S}$ [25], [26]. Here, $\boldsymbol{S}$ is defined as

$$\begin{aligned}\boldsymbol{S} &= \frac{1}{N}\sum_{n=1}^{N}(\boldsymbol{x}_n - \overline{\boldsymbol{x}})(\boldsymbol{x}_n - \overline{\boldsymbol{x}})^H \\ &= \frac{1}{N}\sum_{n=1}^{N}\left(\boldsymbol{x}_n - \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n\right)\left(\boldsymbol{x}_n - \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n\right)^H.\end{aligned} \tag{1}$$

Thus, the dimension reduction or noise reduction can be achieved via reconstructing the observation within the subspace composed of the selected principal components and discarding the residual noise subspace, i.e.,

$$\boldsymbol{s}_n = \sum_{k=1}^{M}(\boldsymbol{x}_n^H\boldsymbol{u}_k)\boldsymbol{u}_k \tag{2}$$

where $\boldsymbol{s}_n$ is the denoised signal and $\boldsymbol{u}_k$ is the eigenvector corresponding to the kth largest eigenvalue of the covariance matrix $\boldsymbol{S}$ defined in (1). Therefore, for the purpose of denoising, the central issue in PCA is to precisely estimate the size of signal subspace, i.e., M.

The size of the signal subspace M reflects the amount of useful information contained in the observations. If M is chosen smaller than the real one, some useful information will be lost. On the contrary, if it is chosen larger than the real one, some fake information will be introduced, which will confuse us in some applications.

In this section, we propose a nonparametric Bayesian PCA model with the BP prior to adaptively determine the size of the signal subspace. The BP-PCA model can be expressed as follows:

$$\boldsymbol{x}_n = \boldsymbol{U}\,\mathrm{Diag}(\mathbf{z})\boldsymbol{U}^H\boldsymbol{x}_n + \boldsymbol{\varepsilon}_n \tag{3}$$

where $\boldsymbol{x}_n \in \mathbb{C}^{D\times 1}$ denotes the time-domain observations; $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_D]$ is the $D \times D$ eigenvector matrix of the observation covariance matrix $\boldsymbol{S}$ defined in (1), with the column vector $\boldsymbol{u}_k$ denoting the eigenvector corresponding to the kth largest eigenvalue λ_k ($k = 1, \ldots, D$ and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D$); $\boldsymbol{\varepsilon}_n \in \mathbb{C}^{D\times 1}$ denotes the noise component; and $\mathrm{Diag}(\mathbf{z})$ denotes a diagonal matrix with the binary vector $\mathbf{z} = [z_1, \ldots, z_D]^T$ being the diagonal elements.

1) Binary Vector: In the binary vector $\mathbf{z}$, $z_k = 1$ means that the corresponding kth eigenvector belongs to the signal subspace and $z_k = 0$ means that the corresponding kth eigenvector belongs to the noise subspace. Thus, we use the Bernoulli prior for z_k, i.e., $z_k \sim Bernoulli(\pi_k)$ with π_k denoting the probability of $z_k = 1$. In order to develop a Bayesian treatment for the problem, we need to introduce a prior distribution $p(\pi_k)$ over the parameter π_k. This distribution $p(\pi_k)$ should satisfy two conditions. 1) It should allow the closed-form Bayesian inference for z_k. 2) In the learning process, it controls that there are only a small number of $\{z_k\}_{k=1}^{D}$ being equal to 1, i.e., imposing sparseness on $\{z_k\}_{k=1}^{D}$.

The proper distribution that satisfies the above two conditions is Beta distribution, i.e., $\pi_k \sim Beta(a_0, b_0)$ with hyperparameters $a_0 > 0$, $b_0 > 0$. Since the density function $Beta(a_0, b_0)$ is the conjugate prior for the Bernoulli variable z_k, where the conjugate prior means that the posterior distribution calculated via the production of the prior and the likelihood function has the same functional form as the prior, this BP prior allows the closed-form Bayesian inference for z_k. In addition, for $\pi_k \sim Beta(a_0, b_0)$, we note that the expectation and variance of π_k can be represented via $\langle \pi_k \rangle = \frac{a_0}{a_0+b_0}$ and $\left\langle (\pi_k - \langle \pi_k \rangle)^2 \right\rangle = \frac{a_0 b_0}{\left[(a_0+b_0)^2(a_0+b_0+1)\right]}$. To impose sparseness on $\{z_k\}_{k=1}^{D}$, we may let $\frac{a_0}{a_0+b_0} \approx 0$, and therefore, it is probable that π_k is close to zero. In the real application, we usually specify $a_0 = \frac{1}{K}$ and $b_0 = \frac{K-1}{K}$ with K denoting a large integer, same as the corresponding priors used in [33] and [34]. Thus, $\langle \pi_k \rangle \approx 0$ with a very small variance $\left\langle (\pi_k - \langle \pi_k \rangle)^2 \right\rangle$.

Based on the above analysis, in this paper, we use the BP prior on the binary vector $\mathbf{z}$

$$z_k \sim Bernoulli(\pi_k), \pi_k \sim Beta(a_o, b_o), \quad k = 1, \ldots, D. \tag{4}$$

2) Noise Component: The noise variable $\boldsymbol{\varepsilon}_n$ is independent of the signal component $\boldsymbol{U}\,\mathrm{Diag}(\mathbf{z})\boldsymbol{U}^H\boldsymbol{x}_n$. In most cases, the noises in the inphase and quadrature echoes of radar targets can be assumed to be Gaussian white noises. Therefore, we assume that the model noise is drawn i.i.d. from a complex Gaussian distribution with the zero-mean. The noise variance or precision is assumed unknown, and is learned within the model inference. Mathematically

$$\boldsymbol{\varepsilon}_n \sim \mathbb{C}N(0, \gamma^{-1}\boldsymbol{I}_D)$$
$$p(\gamma) = Gamma(\gamma | c_0, d_0) = \Gamma(c_0)^{-1} d_0^{c_0} \gamma^{c_0 - 1} e^{-d_0\gamma} \tag{5}$$

where $\mathbb{C}N(\bullet)$ represents the complex Gaussian distribution with $p(\boldsymbol{\varepsilon}_n|\gamma) = \frac{1}{(\pi\gamma^{-1})^D}\exp\left(-\gamma\boldsymbol{\varepsilon}_n^H\boldsymbol{\varepsilon}_n\right)$, $Gamma(\bullet)$ represents the Gamma distribution with hyperparameters c_0 and d_0, and $\Gamma(c_0) = \int_0^\infty t^{c_0-1}e^{-t}dt$ denotes the Gamma function.

Based on the above analysis, we now summarize the BP-PCA model in (6) and show its graphical representation in Fig. 1. In the graph, the circle nodes denote the latent variables, the shaded circle node denotes the observed data, and the square nodes denote the hyperparameters. An arrow indicates dependence between variables, i.e., the conditional distribution of a variable given its parents

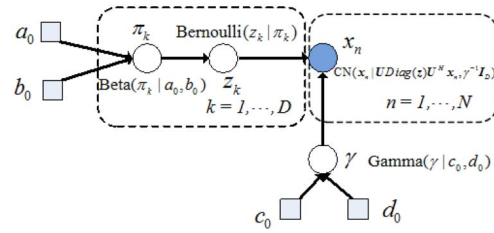

Fig. 1. Graphical representation of the BP-PCA model. The detailed model is described in (6). Here n is the observation sample index, N denotes the number of samples, $\boldsymbol{x}_n$ denotes the nth complex-valued radar observation sample, k is the sample dimension index, and D denotes the total length of each observation sample. The symbols z_k, π_k, and γ are latent variables with the conditional distributions given beside them; a_0, b_0, c_0, and d_0 are hyperparameters. The dashed-line boxes with D and N at the edges of the boxes indicate that there is a stack of such box for the index k and n, respectively.

$$\begin{aligned}
&\boldsymbol{x}_n = \boldsymbol{U}\mathrm{Diag}(z)\boldsymbol{U}^H\boldsymbol{x}_n + \boldsymbol{\varepsilon}_n,\ n = 1, \ldots, N \\
&z_k \sim Bernoulli(\pi_k),\ k = 1, \ldots, D \\
&\pi_k \sim\ Beta\,(a_0, b_0) \\
&\boldsymbol{\varepsilon}_n \sim CN(0, \gamma^{-1}\boldsymbol{I}_D) \\
&\gamma \sim Gamma(c_0, d_0).
\end{aligned} \tag{6}$$

B. VB Inference

In recent decade, VB method is widely used to approximately resolve Bayesian integral [35]. Based on the assumption that parameters and hidden variables are independent of each other, the jointly probability distribution over all parameters and hidden variables can be approximated with a simpler distribution which is a lower bound of original Bayesian integral. The lower bound is increased by optimizing parameters, and the aim is to approximate the real value of original Bayesian integral. In this section, we derive a VB algorithm to perform fast inference for the proposed BP-PCA model shown in (6).

First, we categorize the variables in (6) into three kinds, i.e., observed data $X = \{\boldsymbol{x}_n\}_{n=1}^N$, latent variables $\Phi = \{\boldsymbol{z}, \boldsymbol{\pi}, \gamma\}$, and hyperparameters $\mathbf{H} = \{a_0, b_0, c_0, d_0\}$. The VB inference seeks a variational distribution $q(\Phi)$ to approximate the true posterior distribution of the latent variables $p(\Phi|\boldsymbol{X}, \mathbf{H})$.The expression can be described as follows:

$$\ln p(\boldsymbol{X}|\mathbf{H}) = L(q(\Phi)) + KL(q(\Phi)|p(\Phi|\boldsymbol{X}, \mathbf{H})) \tag{7}$$

where $\ln p(\boldsymbol{X}|\mathbf{H})$ denotes the likelihood of the observations, $KL(q(\Phi)|p(\Phi|\boldsymbol{X}, \mathbf{H}))$ denotes the Kullbac—Leibler divergence between the variational approximation $q(\Phi)$, and the true joint posterior $p(\Phi|\boldsymbol{X}, \mathbf{H})$. Since $KL(q(\Phi)|p(\Phi|\boldsymbol{X}, \mathbf{H})) \geq 0$, it follows that $L(q(\Phi))$ is a rigorous lower bound on $\ln(P(\boldsymbol{X}|\mathbf{H}))$. Therefore, the optimal parameterization of $q(\Phi)$

is obtained by minimizing $KL(q(\Phi)|p(\Phi|\boldsymbol{X},\mathbf{H}))$, i.e., maximizing the similarity between the variational distribution and the true posterior.

The complete log likelihood for the model (6) is given by

$$\begin{aligned}
L(\Phi,\mathbf{H}) &= DN\ln(\gamma) - \gamma\sum_{n=1}^{N}\begin{matrix}(\boldsymbol{x}_n - \boldsymbol{U}Diag(z)\boldsymbol{U}^H\boldsymbol{x}_n)^H\\ \bullet(\boldsymbol{x}_n - \boldsymbol{U}Diag(z)\boldsymbol{U}^H\boldsymbol{x}_n)\end{matrix}\\
&+\sum_{k=1}^{D}\left[\begin{matrix}z_k\ln(\pi_k)+(1-z_k)\ln(1-\pi_k)+\ln\left(\frac{\Gamma(a_0+b_0)}{\Gamma(a_0)\Gamma(b_0)}\right)\\ +(a_0-1)\ln(\pi_k)+(b_0-1)\ln(1-\pi_k)\end{matrix}\right]\\
&+\ln\left(\Gamma(c_0)^{-1}d_0^{c_0}\gamma^{c_0-1}e^{-d_0\gamma}\right)+\text{const}\\
&= DN\ln(\gamma)-\gamma\sum_{n=1}^{N}(\boldsymbol{x}_n^H\boldsymbol{x}_n - \boldsymbol{x}_n^H\boldsymbol{U}Diag(z)\boldsymbol{U}^H\boldsymbol{x}_n)\\
&+\sum_{k=1}^{D}\left[\begin{matrix}z_k\ln(\pi_k)+(1-z_k)\ln(1-\pi_k)+\ln\left(\frac{\Gamma(a_0+b_0)}{\Gamma(a_0)\Gamma(b_0)}\right)\\ +(a_0-1)\ln(\pi_k)+(b_0-1)\ln(1-\pi_k)\end{matrix}\right]\\
&+[-\ln\Gamma(c_0)+c_0\ln d_0+(c_0-1)\ln\gamma - d_0\gamma]+\text{const.}
\end{aligned}\tag{8}$$

To derive the VB update equation for each hidden variable, we need to write the expected complete log likelihood with respect to all other variational posteriors and neglect terms not depending on the derived hidden variable. The derivations are as follows.

1) $q(z)$ Distribution: In order to estimate $q(z)$, we calculate $\langle L(\Phi,\mathbf{H})\rangle_{q(\pi)q(\gamma)}$ from (8) and retain only the terms that contain $z=[z_1,\ldots,z_D]^T$

$$\begin{aligned}\langle L(\Phi,\mathbf{H})\rangle_{q(\pi)q(\gamma)} &= \langle\gamma\rangle\sum_{n=1}^{N}\sum_{k=1}^{D}z_k|w_{nk}|^2+\sum_{k=1}^{D}[z_k\langle\ln\pi_k\rangle\\ &+(1-z_k)\langle\ln(1-\pi_k)\rangle]+\text{const}\end{aligned}\tag{9}$$

with w_{nk} denoting the kth element of $\boldsymbol{w}_n=\boldsymbol{U}^H\boldsymbol{x}_n$.

From (9), we can infer that $q(z)$ is of the form

$$\begin{aligned}q(z)&=\prod_{k=1}^{D}q(z_k)\\ &=\prod_{k=1}^{D}Bernoulli\left(\frac{p(z_k=1)}{p(z_k=1)+p(z_k=0)}\right)\end{aligned}\tag{10}$$

where the probabilities of $z_k=1$ and $z_k=0$ are proportional to

$$\begin{aligned}&p(z_k=1)\propto\exp(\langle\ln\pi_k\rangle)\times\exp\left(\langle\gamma\rangle\sum\nolimits_{n=1}^{N}\|w_{nk}\|^2\right)\\ &p(z_k=0)\propto\exp(\langle\ln(1-\pi_k)\rangle).\end{aligned}\tag{11}$$

The expectation mentioned above can be calculated as follows:

$$\begin{aligned}\langle\ln\pi_k\rangle&=\varphi(\widetilde{a_k})-\varphi(\widetilde{a_k}+\widetilde{b_k})\\ \langle\ln(1-\pi_k)\rangle&=\varphi(\widetilde{b_k})-\varphi(\widetilde{a_k}+\widetilde{b_k})\end{aligned}\tag{12}$$

where $\varphi(\cdot)$ denotes the digamma function; $\widetilde{a_k}$ and $\widetilde{b_k}$ are defined in the update for $\boldsymbol{\pi}$.

Then, the posterior expectation $\langle z_k\rangle$ is

$$\langle z_k\rangle=\frac{1}{1+\exp(\langle\ln(1-\pi_k)\rangle-\langle\ln\pi_k\rangle-\langle\gamma\rangle\sum_{n=1}^{N}\|w_{nk}\|^2)}.\tag{13}$$

2) $q(\boldsymbol{\pi})$ Distribution: In order to estimate $q(\boldsymbol{\pi})$, we calculate $\langle L(\Phi,\mathbf{H})\rangle_{q(z)q(\gamma)}$ from (8) and retain only the terms that contain $\boldsymbol{\pi}=[\pi_1,\ldots,\pi_D]^T$

$$\begin{aligned}&\langle L(\Phi,\mathbf{H})\rangle_{q(z)q(\gamma)}\\ &=\sum_{k=1}^{D}\left[\begin{matrix}z_k\ln(\pi_k)+(1-z_k)\ln(1-\pi_k)\\ +(a_0-1)\ln(\pi_k)+(b_0-1)\ln(1-\pi_k)\end{matrix}\right]\\ &\quad+\text{const.}\end{aligned}\tag{14}$$

From (14), we can infer that $q(\boldsymbol{\pi})$ is of the form

$$q(\boldsymbol{\pi})=\prod_{k=1}^{D}q(\pi_k)=\prod_{k=1}^{D}Beta(\pi_k|\widetilde{a}_k,\widetilde{b_k})\tag{15}$$

where

$$\begin{aligned}\widetilde{a_k}&=a_0+\langle z_k\rangle\\ \widetilde{b_k}&=b_0+1-\langle z_k\rangle.\end{aligned}\tag{16}$$

3) $q(\gamma)$ Distribution: In order to estimate $q(\gamma)$, we calculate $\langle L(\Phi,\mathbf{H})\rangle_{q(z)q(\boldsymbol{\pi})}$ from (8) and retain only the terms that contain γ

$$\begin{aligned}&\langle L(\Phi,\mathrm{H})\rangle_{q(z)q(\boldsymbol{\pi})}\\ &=DN\ln(\gamma)-\gamma\sum_{n=1}^{N}\begin{matrix}(\boldsymbol{x}_n^H\boldsymbol{x}_n\\ -\boldsymbol{x}_n^H\boldsymbol{U}\langle Diag(z)\rangle\boldsymbol{U}^H\boldsymbol{x}_n)\end{matrix}\\ &\quad+(c_0-1)\ln\gamma+\text{const.}\end{aligned}\tag{17}$$

From (17), we can infer that $q(\gamma)$ is of the form

$$q(\gamma)=Gamma(\gamma\,|\,\widetilde{c},\widetilde{d})\tag{18}$$

where

$$\begin{aligned}\widetilde{c}&=c_0+DN\\ \widetilde{d}&=d_0+\sum\nolimits_{n=1}^{N}(\boldsymbol{x}_n^H\boldsymbol{x}_n-\langle z^T\rangle\left(|\boldsymbol{w}_n|^2\right).\end{aligned}\tag{19}$$

Then, the posterior expectation of $\langle\gamma\rangle$ is

$$\langle\gamma\rangle=\frac{\widetilde{c}}{\widetilde{d}}.\tag{20}$$

4) Stopping Criterion: Following each iteration, $q(\Phi)$ is updated and we can obtain the lower bound after the $iter$th iteration $L(q(\Phi))_{iter}=\int q(\Phi)\ln\frac{p(X,\Phi)}{q(\Phi)}d\Phi$, if the change between the $iter$th and the $(iter+1)$th lower bound, i.e., $change=\frac{L(q(\Phi))_{iter+1}-L(q(\Phi))_{iter}}{L(q(\Phi))_{iter}}$ falls below a predefined threshold (e.g., 10^{-6}), the algorithm stops.

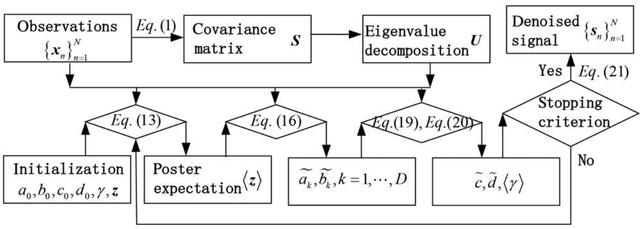

Fig. 2. Flowchart of the proposed denoising method.

TABLE I
DETAILED ITERATION PROCEDURE FOR BP-PCA

Algorithm 1: VB based inference for BP-PCA
Step 1: Initialize the parameters $a_0, b_0, c_0, d_0, \gamma, z$.
Step 2: Calculate the covariance matrix S of the time-domain observations $\{x\}_{n=1}^N$ via Equation (1).
Step 3: Obtain the eigenvector matrix $U = [u_1, \cdots, u_D]$ via the eigenvalue decomposition of S with the column vector u_k denoting the eigenvector corresponding to the kth largest eigenvalue λ_k ($k = 1, \cdots, D$ and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D$).
Step 4: Update the $q(z)$ distribution via Equation (13) with the initial parameters in Step 1, the time-domain observations $\{x\}_{n=1}^N$ and the eigenvector matrix $U = [u_1, \cdots, u_D]$ obtained in Step 3.
Step 5: Update the $q(\pi)$ distribution via Equation (16) with the posterior expectation $\langle z \rangle$ obtained in Step 4.
Step 6: Update the $q(\gamma)$ distribution via Equations (19) and (20) with the initial parameters obtain in Step 1, the time-domain observations $\{x\}_{n=1}^N$, the eigenvector matrix $U = [u_1, \cdots, u_D]$ obtained in Step 3, the posterior expectation $\langle z \rangle$ obtained in Step 4 and the parameters $\tilde{a}_k, \tilde{b}_k$ obtained in Step 5.
Step 7: If the stopping criterion described in Section II-B is satisfied, Equation (24) can be utilized to calculate the denoised time-domain signal with the updated parameters described above. If the stopping criterion is not satisfied, return to Step 4 to update the parameters until the stopping criterion is satisfied.

At the convergence of the parameter estimation procedure, we can obtain the posterior expectation $\langle z \rangle$. In practice, during the iterations, we generally find that each $\langle z_k \rangle$ tends to 0 or 1 automatically. Then, we can get the denoised signal as

$$\boldsymbol{s}_{\mathbf{n}} = \boldsymbol{U} \operatorname{diag}(\langle \mathbf{z} \rangle) \boldsymbol{U}^H \boldsymbol{x}_n. \tag{21}$$

The experimental results show that our BP-PCA model can automatically select the eigenvectors corresponding to the large eigenvalues. Thus, (21) accords with (2), whereas the size of the signal subspace M can be automatically learned from the observations via our BP-PCA.

We summarize the above iteration procedure in Fig. 2 and Table I.

C. Simulation Example

We design a simulation example in this section to verify the effectiveness of the proposed algorithm. We also compare the proposed method with two denoising methods proposed in [5] and [29]. For simplicity, in the following experiments, the proposed BP-PCA method, the denoising method proposed in [5], and the denoising method proposed in [29] are referred to as BP-PCA, CLEAN, and CPPCA-AIC, respectively.

Assume that the observation is composed of complex sinusoidal signals and noise component, and the signal model of which can be written in the following form:

$$x(t) = s'(t) + \varepsilon(t) = \sum_{q=1}^{M} A_q \exp(j2\pi f_q t) + \varepsilon(t) \tag{22}$$

where $x(t)$, $s'(t)$, and $\varepsilon(t)$ denote the observation, the signal without noise, and the zero-mean complex Gaussian white noise component, respectively. In (22), M denotes the number of complex sinusoidal signals, i.e., the size of signal subspace; A_q and f_q are the amplitude and frequency of the qth sinusoidal signal with $q = 1, \ldots, M$. In our experiment, we set the size of the signal subspace $M = 6$. The amplitudes of the six complex sinusoidal signals are 1, 2, 0.5, 0.5, 3.3, and 0.8, respectively, with the corresponding frequencies −1200, −800, −300, 400, 900, and 1400 Hz. Based on the signal model shown in (22), we simulate $N = 545$ observation samples with the length of each sample $D = 256$. In this example, the signal-noise ratio (SNR) of the observation samples is 0 dB with the SNR defined as follows:

$$SNR = 10 \times \log_{10} \frac{\overline{P}_{s'}}{P_{\varepsilon}} = 10 \times \log_{10} \frac{\frac{1}{D} \sum_{t=1}^{D} P_{s'(t)}}{P_{\varepsilon}} \tag{23}$$

where $\overline{P}_{s'}$ denotes the average power of signal component, $P_{s'(t)} = |s'(t)|^2$ denotes the power of the tth element of the signal component, and P_{ε} denotes the power of noise.

In this experiment, the initialization of parameters in (6) is $K = 2000$, $a_0 = b_0 = 1$, and $c_0 = d_0 = 10^{-6}$.

Fig. 3 shows the denoising results of the three methods. Fig. 3(a) and (b) denotes the Doppler spectrum of the signal component $s'(t)$ and the Doppler spectrum of the observation $x(t)$, respectively. Fig. 3(c)–(e) shows the Doppler spectra of the denoised signal $s(t)$ obtained via BP-PCA, CLEAN, and CPPCA-AIC, respectively. Usually, for the noise reduction task in the real classification application, the desired result is not only to remove the noise component but also to preserve the structure of the Doppler spectrum. From Fig. 3, we can note that the proposed BP-PCA method can achieve good denoising performance while preserving the structure of signal component $s'(t)$ well. On the contrary, although CLEAN and CPPCA-AIC can reduce the noise component in the signal, they also change the structure of signal component. Fig. 4 shows the final value of $\langle z_k \rangle$ with $k = 1, \ldots, D$ in BP-PCA, from which we can see that the proposed BP-PCA model can automatically select the eigenvectors corresponding to the largest six eigenvalues.

In order to obtain a quantitative evaluation of the noise reduction performance, we define the MSE between the denoised signal and the original signal under high SNR as follows:

$$\text{MSE} = \frac{\|S - S'\|_2}{\|S'\|_2} \tag{24}$$

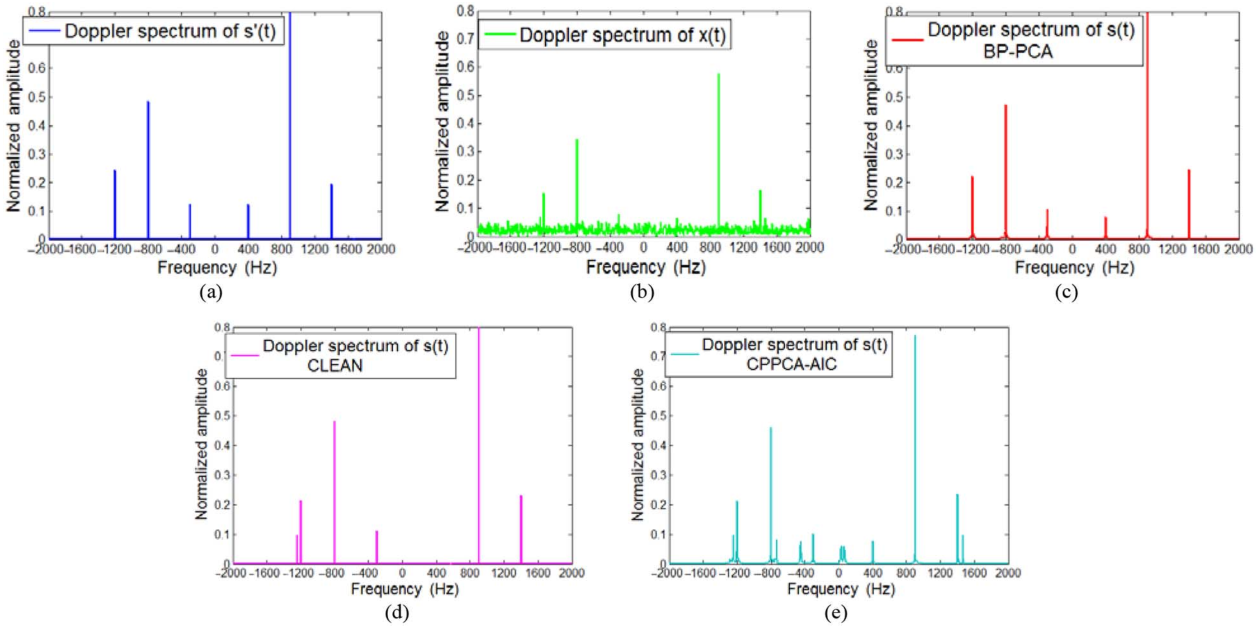

Fig. 3. Noise reduction result on simulated data. (a) Doppler spectrum of the signal component $s'(t)$. (b) Doppler spectrum of the observation $x(t)$. (c) Doppler spectrum of the denoised signal $s(t)$ via BP-PCA. (d) Doppler spectrum of the denoised signal $s(t)$ via CLEAN. (e) Doppler spectrum of the denoised signal $s(t)$ via CPPCA-AIC.

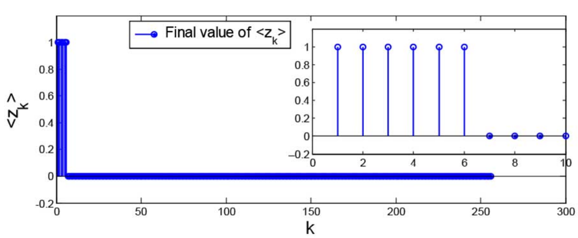

Fig. 4. Final value of $\langle z_k \rangle$ $(k = 1, \ldots, D)$ in BP-PCA for this simulation example. The ground truth is $M = 6$.

where $\|\bullet\|_2$ denotes the l_2-norm of the signal, $S = |\mathrm{FFT}(s)|$ and $S' = |\mathrm{FFT}(s')|$ with $\boldsymbol{S}$ and s' denote the denoised signal and the original signal under high SNR, respectively, and the function $\mathrm{FFT}(\bullet)$ denotes the fast Fourier transformation (FFT).

Fig. 5 shows the average MSE of the simulated samples. In Fig. 5, "Noisy" denotes the average MSE between the noisy signals and the original signals under the high SNR; "BP-PCA" denotes the average MSE between the denoised signals via BP-PCA-based noise reduction method and the original signals; "CLEAN" denotes the average MSE between the denoised signals via CLEAN-based noise reduction method [5] and the original signals; "CPPCA-AIC" denotes the average MSE between the denoised signals via CPPCA-AIC-based noise reduction method [29] and the original signals. We can see that the proposed method achieves the minimum average MSE, especially in the low SNR cases with $\mathrm{SNR} = 0 - 25$ dB. In order to visually compare the MSEs of the three denoising methods more clearly, we replot the lines of "BP-PCA," "CLEAN," and "CPPCA-AIC" in Fig. 5(b).

III. Micro-Doppler Signature Analysis and Denoising Results on Measured Data

A. Introduction of the Measured Data

In this paper, the measured data are collected from two kinds of targets, i.e., 1) airplane targets including turbojet aircraft, prop aircraft and helicopter, and ground moving targets including a moving vehicle; and 2) one or two walking persons.

Our measured data from airplane targets are collected under the cooperative measurement experiments with a low-resolution radar, which works at L-band with a carrier frequency of 1.3 GHz; therefore, the high SNR can be guaranteed and $\mathrm{SNR} \approx 40$ dB. One simple way to calculate the SNR of the measured signal is briefly introduced as follows. We first use the CLEAN technique [5], [36] to extract the target signal component and the noise component. Then, the corresponding power of target signal component and the power of noise component can be obtained. Finally, we can calculate the approximate SNR of the measured data via (23). The dwell time is 200.8 ms and the pulse repetition frequency (PRF) is 4 KHz. For the airplane data, we have totally 239 measured samples of which 62 samples are from turbojet aircrafts, 96 samples are from prop aircrafts, and 81 samples are from helicopters.

Our ground moving target data are measured from two kinds of targets, i.e., 1) a moving vehicle and 2) one or two walking persons. The low-resolution radar for collecting the data works at Ka-band with a carrier frequency of 28.7 GHz, its dwell time is 1.2 s, and the PRF is 3 kHz. For the moving vehicle data, we have totally 100 measured samples from a wheeled vehicle which recedes or approaches radar. The moving vehicle data are obtained under the low SNR condition ($\mathrm{SNR} \approx 15 - 18$ dB). The walking person (persons) data are collected under two experimental scenarios. 1) A single person walks toward radar,

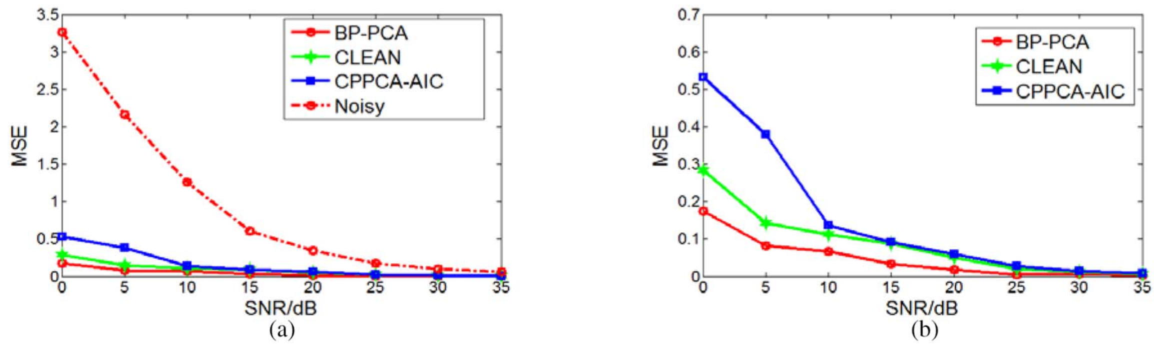

Fig. 5. Variation of the average MSEs of the simulated data with the SNR conditions via the different noise reduction methods. (a) Average MSEs between the noisy signals and original signals, denosied signals via BP-PCA and original signals, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively. (b) Average MSEs between denoised signals via BP-PCA and original signal, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively.

where we have totally 120 samples. 2) Two persons walk toward or away from radar, their walking speeds are close but not the same, where we have totally 80 samples. These data from walking person (persons) are obtained under the high SNR condition (SNR ≈ 35 dB).

In Sections III-B and III-C, we first analyze the different micro-Doppler signatures of different kinds of targets. Then, the proposed noise reduction method is applied to the measured data to evaluate its noise reduction performance with comparison with the two related noise reduction methods proposed in papers [5] and [29]. Since our measured radar data from airplane targets and walking person (persons) are collected under the high SNR condition, we add simulated zero-mean complex Gaussian white noises to them to quantitatively evaluate the denoising performance. For the measured data from the moving vehicle, we denoise them directly using the proposed method since they are collected under the low SNR condition.

B. Experiments on Measured Data From Airplane Targets

Fig. 6(a)–(c) shows the Doppler spectra of the measured samples from one turbojet aircraft, one prop aircraft, and one helicopter, respectively. In Fig. 6, the largest Doppler frequency corresponds to the fuselage component, while the rest of periodic Doppler spectrum lines are induced by the rotating parts in the aircraft. We can see that the micro-Doppler signatures of the three kinds of targets are very discriminated, i.e., a turbojet aircraft has a spectrum with the number of spectrum lines much fewer than that of a prop aircraft and the interval between those lines much larger than that of a prop aircraft, and a prop aircraft has a spectrum with the number of spectrum lines much fewer than that of a helicopter and the interval between the spectrum lines much larger than that of a helicopter which is in accordance with the theoretical analysis in [5] and [37]. In practice, the micro-Doppler component of a turbojet aircraft cannot always be seen due to the shelter of the blades by the turbine duct.

1) Denoising Results: Our measured data from airplane targets are collected under the cooperative measurement experiments with a low-resolution radar; therefore, the high SNR can be guaranteed and SNR ≈ 40 dB. In order to test the noise

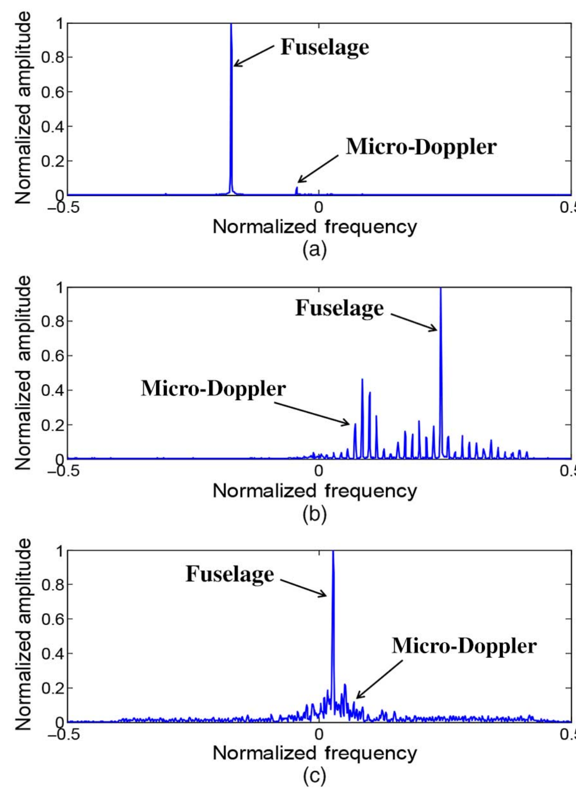

Fig. 6. Doppler spectra of the measured radar data from the three kinds of aircraft under the high SNR condition is about 40 dB. (a) Turbojet aircraft. (b) Prop aircraft. (c) Helicopter.

reduction performance of the proposed BP-PCA method, we add simulated zero-mean complex Gaussian white noises to the measured data. Here, we should point out that the SNR used in this paper is defined in (23), where $\overline{P}_{s'}$ in the following experiments means the signal power averaged within the dwell time for a radar echo after pulse compression. Thus, here the SNR refers to the average SNR of a single pulse rather than the SNR of multiple pulses after coherent or noncoherent integration.

Fig. 7 gives a noise reduction example based on a measured sample from a prop aircraft. Fig. 7(a) and (b) shows the Doppler spectra of the original signal under the high SNR condition and that under the SNR condition of 10 dB, respectively. The

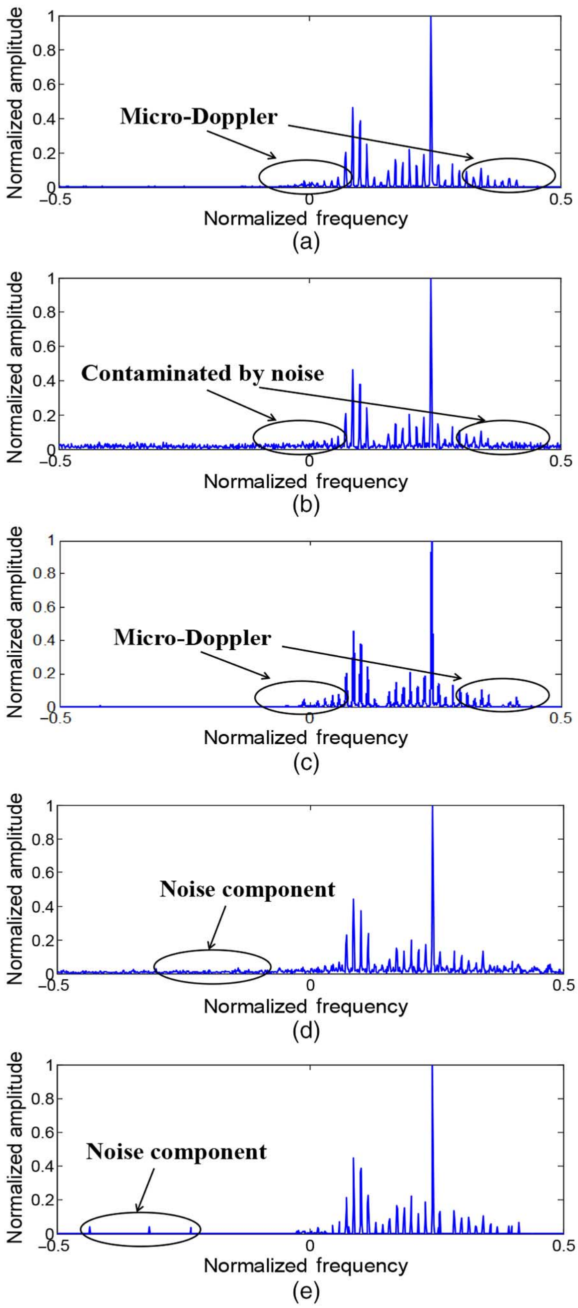

Fig. 7. Doppler spectrum of a measured sample from a prop aircraft. The measured data sample used here is the same as that in Fig. 6(b). (a) Doppler spectrum of the original signal under the high SNR condition is about 40 dB. (b) Doppler spectrum of the noised signal under the SNR condition of 10 dB. (c) Doppler spectrum of the denoised signal via BP-PCA. (d) Doppler spectrum of the denoised signal via CPPCA-AIC. (e) Doppler spectrum of the denoised signal via CLEAN.

Doppler spectra of the corresponding denoised signals via the proposed BP-PCA-, CPPCA-AIC-, and CLEAN-based noise reduction approaches from the noised signal in Fig. 7(b) are depicted in Fig. 7(c)–(e), respectively. Comparing Fig. 7(a)–(e), we can see that the proposed noise reduction approach can not only effectively reduce the noise component but also have little effect on the original signal component, while there are still some noise components left in the denoised signals via CPPCA-AIC- and CLEAN-based methods.

In order to visually compare the noise reduction performances more clearly, the corresponding time–frequency (TF) spectrograms are given in Fig. 8. The signals in Fig. 8(a)–(e) are same as those in Fig. 7(a)–(e), respectively.

Fig. 9(a) shows the average MSE of the 239 samples from airplane targets, which includes 62 samples from turbojet aircrafts, 96 samples from prop aircrafts, and 81 samples from helicopters. The MSE is calculated using (24). In Fig. 9(a), "Noisy" denotes the average MSE between the noisy signals and the original signals under the high SNR about 40 dB; "BP-PCA" denotes the average MSE between the denoised signals via BP-PCA-based noise reduction method and the original signals; "CLEAN" denotes the average MSE between the denoised signals via CLEAN-based noise reduction method [5] and the original signals; "CPPCA-AIC" denotes the average MSE between the denoised signals via CPPCA-AIC-based noise reduction method [29] and the original signals. We can see that the proposed method achieves the minimum average MSE between the Doppler spectra of the denoised signals and those of the original signals, especially in the low SNR cases with $\mathrm{SNR} = 5 - 20$ dB. In order to visually compare the MSEs of the three denoised methods more clearly, we replot the lines of "BP-PCA," "CLEAN," and "CPPCA-AIC" in Fig. 9(b).

C. Experiments on Measured Data From Ground Moving Targets

1) Analysis of Micro-Doppler Signatures of Ground Moving Targets: In this paper, our ground moving target data are measured from two kinds of targets, i.e., 1) a moving vehicle and 2) one or two walking persons. The low-resolution radar for collecting the data works at Ka-band with a carrier frequency of 28.7 GHz, its dwell time is 1.2 s, and the PRF is 3 KHz. The measured data from the walking person (persons) are obtained under the SNR condition of about 35 dB, while the measured data from the moving vehicle are collected under the SNR condition of about 15–18 dB due to the large distance between radar and target.

When a person walks or a vehicle moves, the micromotion will be generated by the swinging of arms and legs or the rolling of the track. The subechoes induced by the swinging arms and legs are time-varying function like $A\sin(\omega t + \theta)$ with A denoting the amplitude of the swing, ω denoting the frequency of the swing, and θ denoting the initial phase of the swing. Thus, the radar return from a walking person is time-varying and periodical, where its amplitude and cycle vary from person to person since they depend on the swinging amplitudes and frequencies of the person's arms and legs. On the contrary, the micro-Doppler components induced by the rolling of the track are not time varying and they are within the Doppler frequency range 0 and $\frac{2v}{\lambda}$ with v and λ denoting the velocity of the bulk motion along the LOS and the wavelength of the transmitted signal, respectively [16].

Since the TF analysis depicts how the instantaneous frequency of a signal varies with time, it is a good tool to analyze the micro-Doppler signatures of a target. Fig. 10 shows the TF spectrograms of the measured data from a moving vehicle,

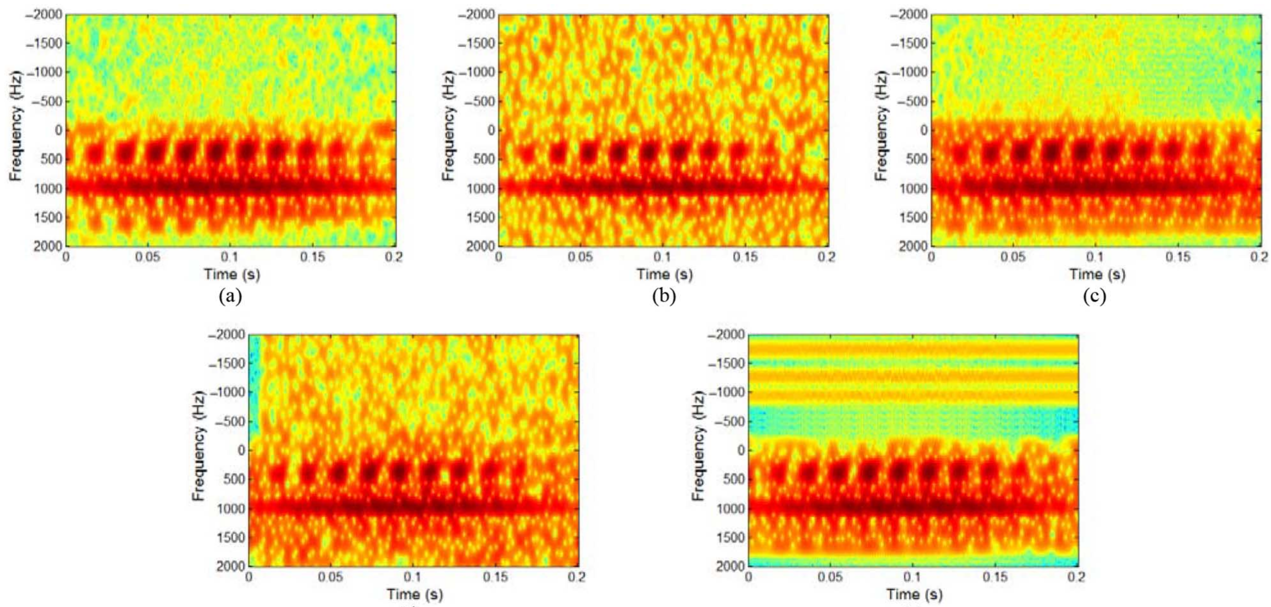

Fig. 8. TF spectrograms of the signals in Fig. 7. (a) TF spectrogram of the signal shown in Fig. 7(a) which is under SNR = 40 dB. (b) TF spectrogram of the signal shown in Fig.7(b) which is under SNR = 10 dB. (c) Denoised result from (b) via BP-PCA. (d) Denoised result from (b) via CPPCA-AIC. (e) Denoised result from (b) via CLEAN-based method.

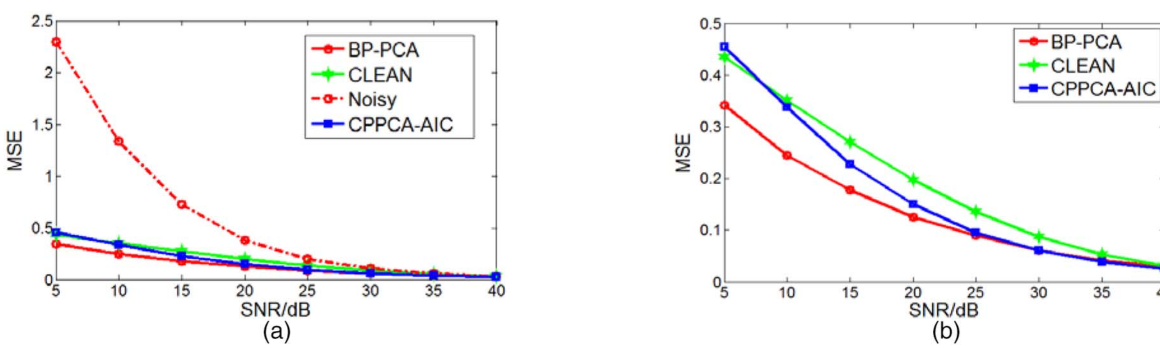

Fig. 9. Variation of the average MSEs of the measured airplane data with the SNR conditions via the different noise reduction methods. (a) Average MSEs between the noisy signals and original signals, denosied signals via BP-PCA and original signals, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively. (b) Average MSEs between denoised signals via BP-PCA and original signal, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively.

one walking person and two walking persons. Here, we use the short-time Fourier transformation (STFT) to obtain the TF spectrogram.

As shown in Fig. 10, the TF spectrograms show rather distinctive characteristics. In Fig. 10(a) and (b), the instantaneous frequency component, which covers a stable frequency band of about 700–800 Hz through the whole dwell time, corresponds to the torso component; while the periodic micro-Doppler modulation components surrounding the torso return come from arm and leg movements. We can see that the more the persons are in an experiment, the more abundant the micro-Doppler components are in the TF spectrogram. This is due to the fact that the lengths of the arm and leg, their swinging frequencies, and movement ranges vary much from person to person. That is to say, the instantaneous frequencies of different persons walking together can hardly overlap totally; thus, the micro-Doppler components from two walking persons are more abundant than those from a single walking person.

We indicate the bulk-motion, micromotion, upper-track micromotion, noise components, and clutter components in Fig. 10(c) and (d) for the data of moving vehicle. We can see that the instantaneous micro-Doppler components of a moving vehicle do not show such periodical property as shown in Fig. 10(a) and (b). The $2v$-Doppler component is induced by the upper-track micromotion. Comparing Fig. 10(c) and (d), we can see that the $2v$-Doppler component cannot always be seen due to the shelter of the upper-track in practice. In addition, as aforementioned, our measured data from walking persons are obtained under high SNR (SNR ≈ 35 dB), while measured data from a moving vehicle are obtained under relatively low SNR (SNR $\approx 15 - 18$ dB).

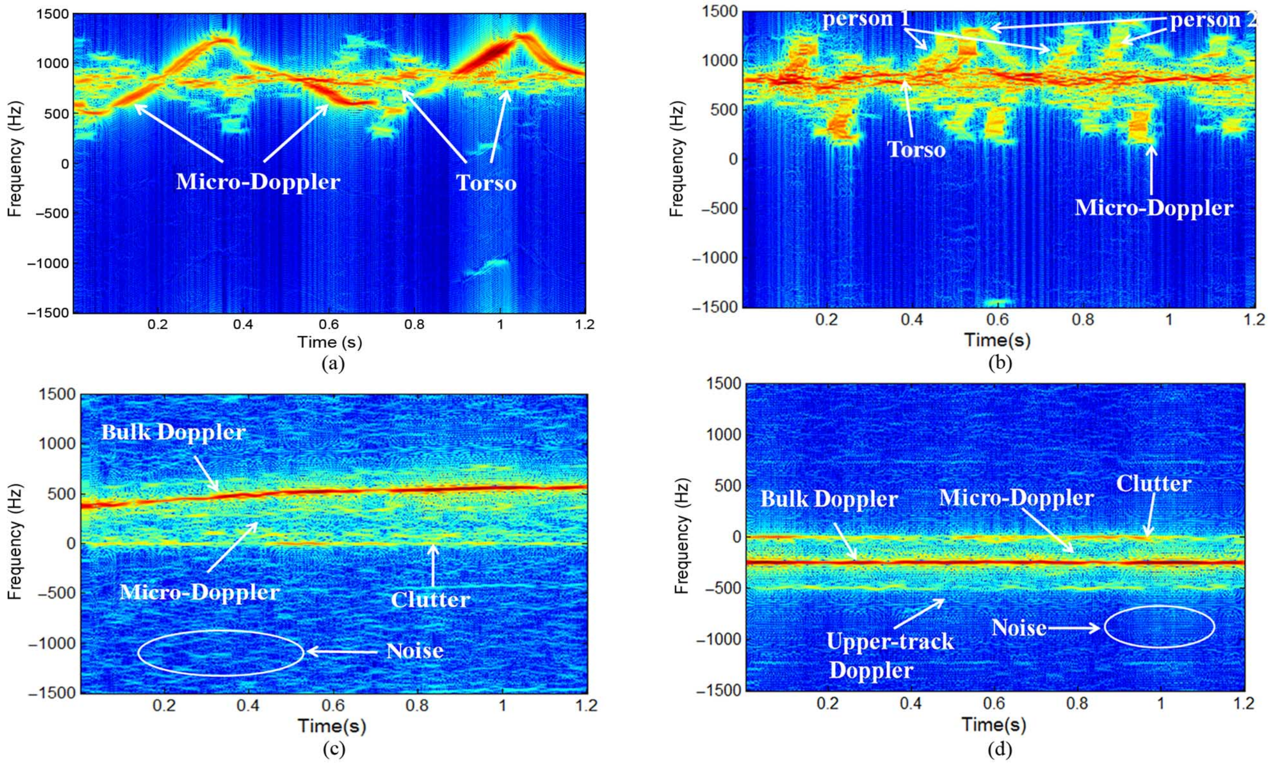

Fig. 10. TF spectrograms of measured echoes from walking human individuals and moving vehicle. (a) One walking person under high SNR (SNR $\approx$ 35 dB). (b) Two walking persons under high SNR (SNR $\approx$ 35 dB). (c) Moving vehicle under low SNR (SNR $\approx$ 15 dB). (d) Moving vehicle under low SNR (SNR $\approx$ 18 dB).

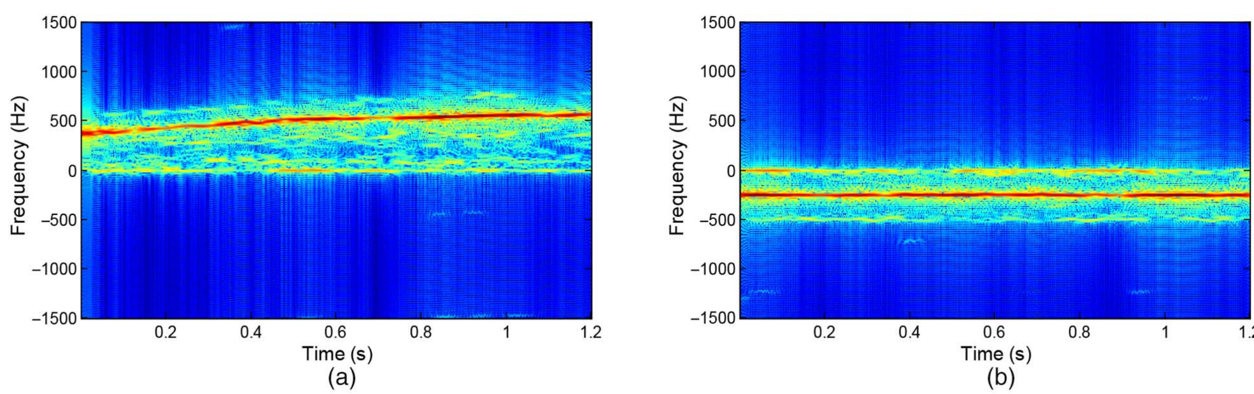

Fig. 11. Denoised TF spectrograms of the measured data from a moving vehicle. (a) Denoised result from Fig. 10(c) via BP-PCA. (b) Denoised result from Fig. 10(d) via BP-PCA.

2) Denoising Results: Fig. 11 shows the noise reduction results of the measured data shown in Fig. 10(c) and (d) using the proposed BP-PCA method. We can see that the proposed BP-PCA-based noise reduction method can work well on the measured noise.

Fig. 12 further compares the noise reduction performance of the proposed BP-PCA method with the two related methods, i.e., 1) CPPCA-AIC method and 2) the CLEAN method. Fig. 12(a) and (b) shows the noised TF spectrograms of the measured data from one to two persons, respectively, both of which are under the SNR condition of 10 dB. The corresponding TF spectrograms under the high SNR condition of about 35 dB are shown in Fig. 10(a) and (b). Fig. 12(c), (e), and (g) shows the denosied results from Fig. 12(a) via the proposed BP-PCA method, the CPPCA-AIC method and the CLEAN method, respectively. Fig. 12(d), (f), and (h) shows the denosied results from Fig. 12(b) via the proposed BP-PCA method, the CPPCA-AIC method, and the CLEAN method, respectively. Comparing Figs. 10 and 12, we can see that the proposed BP-PCA method cannot only reduce the noise component well but also preserve the original target signal, while both the CPPCA-AIC method and the CLEAN method cannot reduce the noise component well, i.e., there are some noise components left in the TF spectrogram after being denoised via the two methods.

Fig. 13(a) shows the average MSE of the 200 samples from walking person (persons), which includes 120 samples from one walking person and 80 samples from two walking persons.

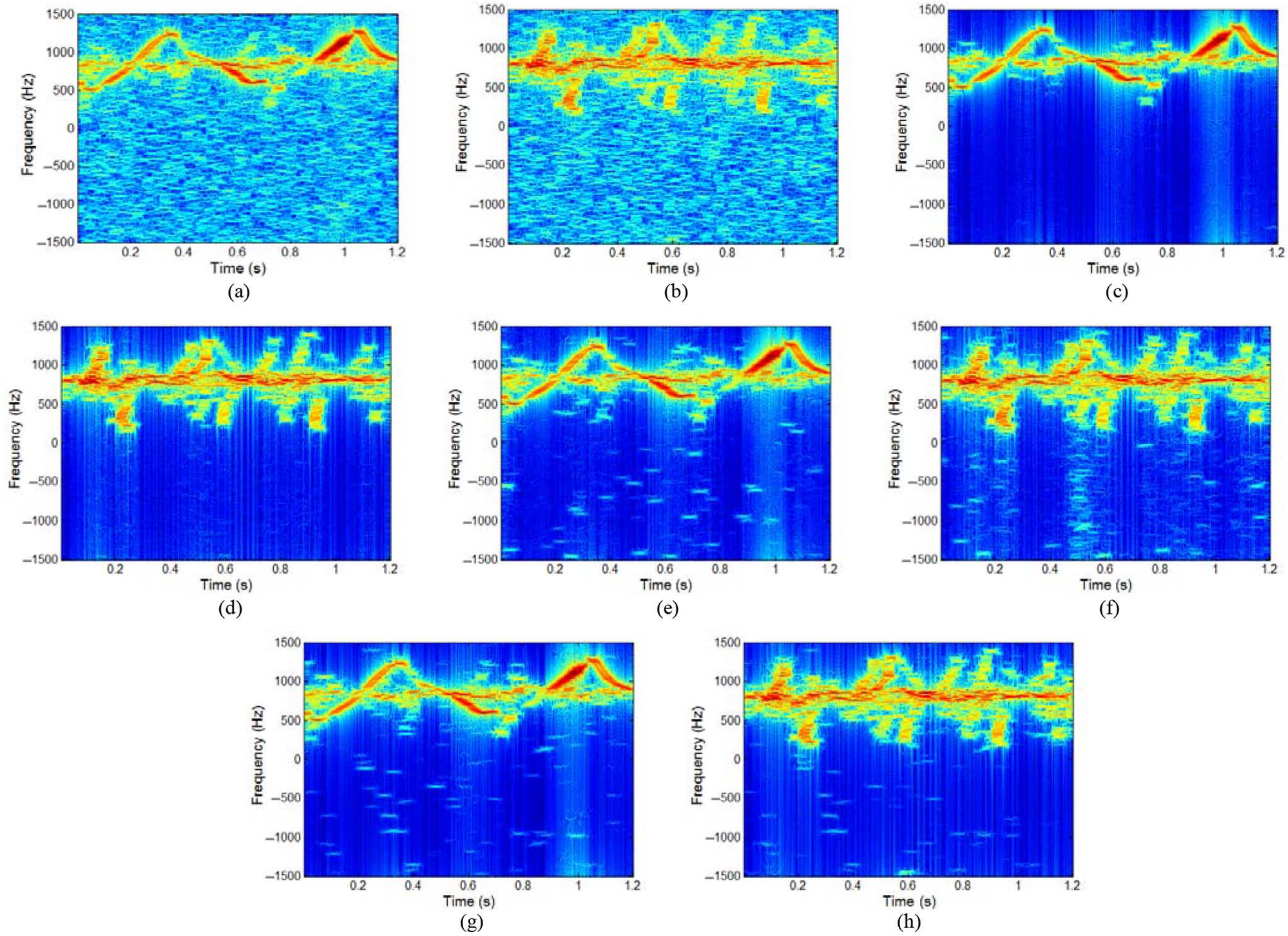

Fig. 12. Noisy and denoised TF spectrograms of the measured data from walking persons. (a) One walking person under SNR $= 10$ dB. (b) Two walking persons under SNR $= 10$ dB. (c) denoised result from (a) via BP-PCA. (d) Denoised result from (b) via BP-PCA. (e) Denoised result from (a) via CPPCA-AIC. (f) Denoised result from (b) via CPPCA-AIC. (g) Denoised result from (a) via CLEAN. (h) Denoised result from (b) via CLEAN. The original TF spectrograms under the high SNR condition of about 35 dB are shown in Fig. 10(a) and (b).

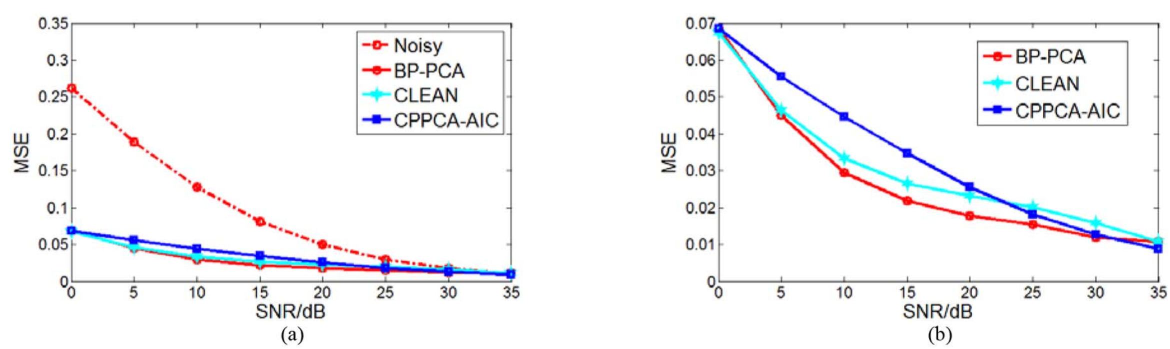

Fig. 13. Variation of the average MSEs of the measured person data with the SNR conditions via the different noise reduction methods. (a) Average MSEs between the noisy signals and original signals, denosied signals via BP-PCA and original signals, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively. (b) Average MSEs between denosied signals via BP-PCA and original signals, denoised signals via CLEAN and original signals, denoised signals via CPPCA-AIC and original signals, respectively.

The MSE is calculated the same way as described in (24) except that we use the STFT to replace the FFT, i.e., here $S = |STFT(s)|$ and $S' = |\mathrm{STFT}(s')|$ with $\boldsymbol{S}$ and s' denoting the denoised signal and the original signal under high SNR, respectively. In Fig. 13(a), "Noisy" denotes the average MSE between the noisy signals and the original signals under the SNR condition of about 35 dB; "BP-PCA," "CPPCA-AIC," and "CLEAN" denote the average MSE between the denoised signals via the corresponding denoising method and the original signal, respectively. We can note that the proposed method achieves the minimum average MSEs between the TF spectrograms of the denoised signals and the TF spectrograms of the original signals, especially in the low SNR cases with $\mathrm{SNR} = 5 - 20$ dB. In order to visually compare the MSEs of the three denoised methods more clearly, we replot the lines of "BP-PCA," "CLEAN," and "CPPCA-AIC" in Fig. 13(b).

IV. Conclusion

A novel efficient method to denoise the returned micro-Doppler signals from airplane targets and ground moving targets with low-resolution radar is proposed in this paper. We first develop a nonparametric Bayesian model for the PCA problem with the BP prior. Then, the BP-PCA model is applied to the low-resolution radar returns for automatic choice of the number of principal components. Noise reduction is accomplished via reconstructing the echo within the subspace composed of the selected principal components. Experimental results based on measured data show that the proposed method cannot only achieve the good noise reduction performances but also preserve the micro-Doppler signatures of the radar returns under the relatively low SNR conditions.

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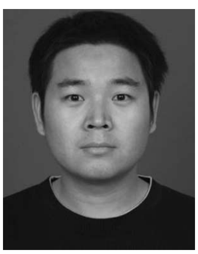

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